

Secured Electronic voting system using Blockchain Technology

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Abstract - Electronic voting has become an increasingly popular topic of discussion in recent years but it has some security issues. Ethereum's smart contract capabilities make it possible to develop tamper-proof and transparent e-voting systems. Smart contracts are self-executing programs that are stored on the Ethereum blockchain, and can be used to automate the process of voting and ensure that the results are accurate and verifiable. This paper explores the use of Ethereum for e-voting, including the advantages and challenges of using this platform for voting. The paper discusses the use of smart contracts for implementing e-voting systems, the security and privacy considerations that must be taken into account, and the potential benefits of using Ethereum for e-voting. The paper also discusses the challenges that must be overcome to ensure the widespread adoption of Ethereum-based e-voting systems, including issues related to scalability, interoperability, and regulatory compliance. In conclusion, this paper argues that Ethereum has the potential to revolutionize the field of e-voting, offering a secure, transparent, and efficient solution for conducting political elections. While there are still many challenges to overcome, the use of Ethereum for e-voting represents an exciting opportunity for governments, businesses, and individuals around the world to participate in democratic processes in a more secure and efficient way.

Keywords - Machine Learning, Data Science, phishing website, Blockchain, Electronic voting

I. INTRODUCTION

Without a shadow of a question, the groundbreaking idea of the blockchain, which is the underlying technology behind the popular cryptocurrency Bitcoin and its successors, is ushering at the beginning of a new age for the Internet and online businesses. While most people's attention is on cryptocurrencies, many previously offline administrative tasks, finance procedures, and everyday activities are now being safely relocated to the Internet as online services. The Implementation of smart contracts, as seen on the Ethereum Space, makes it a potential medium for the global digitalisation of mundane services. Integrating smart contracts onto the blockchain and having them run as planned throughout each round of blockchain updates is a key function of this technology. On the other hand, electronic voting is another hot subject that's also crucial to the future of internet businesses. An attractive option for

creating trustworthy, inexpensive, secure, transparent, and user-friendly electronic voting systems is the blockchain coupled with smart contracts. Due to its reliability, popularity, and availability of smart contract logic, Ethereum and its network are among the best options. It is important for an electronic voting system to be trustworthy in that it prevents voters from casting the same ballot more than once, is completely open and public, and keeps voters personal information confidential. Here, we show how to use Ethereum wallets and the Solidity programming language to build a working smart contract for example electronic voting application on the Ethereum network. People who don't have an Ethereum wallet may still be able to cast a vote by using the Android platform.

In addition to requiring a lot of paperwork, human resources, and time, today's voting methods like ballot box voting and electronic voting are vulnerable to a broad range of security risks including DDoS assaults, polling booth capture, vote tampering and manipulation, and malware attacks. Consequently, people tend to lose faith in the current order of things. Voting data will be stored in the Ethereum wallet after an election has concluded. Every node in the Ethereum Space must agree on the results of a vote cast by a user on an Android device or via an Ethereum wallet. In this work, we present a new approach for a trust-free, distributed voting platform using Blockchain technology to eliminate the need for centralized authority.

In this age of industry 4.0, blockchain technology is becoming increasingly popular. Its high security and openness have led to its widespread adoption in a variety of applications, including voting systems, supply chain management software, healthcare payment systems, commercial enterprise, the Internet of Things, and more. A distributed, unchangeable database, blockchain allows businesses to record transactions and keep track of assets. The innovation brought about by blockchain technology is that it provides a safe and secure method of recording data and building trust among those involved along with a neutral third party. According to the findings, the method is workable and might pave the way for the creation of optimal settings for such an adventure. Data in a blockchain database is stored in a series of blocks. The data is

chronologically consistent because the chain cannot be deleted or altered without network consensus. Consequently, you can create an unalterable or immutable ledger for tracking orders, payments, accounts, and other transactions using blockchain technology. The system is equipped with mechanisms to prevent unauthorized transaction entries and ensure consistency in the shared view of these transactions.

II. EXISTING TECHNOLOGY

The technology used here to crack the problem is Blockchain-Based Computing. Since the initiation of familiar and acquainted initiatives in cryptocurrencies. Bitcoin was the first digital currency to employ Cryptocurrency transactions. Ethereum brought on smart contracts that highlight the benefit and power and elevated results while providing a Stable cryptocurrency solution to Cryptocurrency Transactions. Nick Szabo has Acquainted the speculations of Smart contracts dating back to the 90s, elaborating them as "a set of promises specified in digital form, including protocols within which the parties perform on these promises". A contract in Ethereum is a Segment of code that is to be deployed to the network and is accessed by the Public. The result for this category of code is segmented and validated effectively.

Currently, we refer to a Blockchain-based computational system as a set of technologies constituting the blockchain data structure, a distributed Evaluation, Key cryptography, and smart contracts. This section describes these technologies in greater depth. A peer-to-peer network repeats a series of blocks created by blockchain technology. A block includes the Merkle structured tree header block as well as multiple transactions. Cryptography, being a very sophisticated technological division uses both computer science and mathematics to derive the Data and implement Security. It enables the secure transmission of encrypted and decrypted data over an insecure network. Blockchain technology rectified deficiencies in the current method of elections by making the mechanism of polling transparent and accessible, preventing illicit voting, emboldening data security and verifying the polling results.

This legacy is Satoshi Nakamoto's [1] complete understanding of blockchain and how it works. Indonesia held simultaneous presidential and parliamentary elections on April 17, 2019. This election had at least four major issues: logistics distribution, count of ballots that took a very crucial time, irregular regulation of vote counting, and vote recapitulation errors. Blockchain technology may solve those issues.

[2] In this work, they have applied the method, but for several other use cases, and we have used it as a baseline to develop a peer-to-peer transaction system in lieu of a decentralized voting system.

[3] In the current time, there has been a phenomenal boost in the number of wireless technologies, smart gadgets, and sensors. IT has been anticipated that trillions upon Billions of gadgets will be associated with each other in the not-too-distant future. Therefore, in order

to accommodate such a large number of devices, network designs need to be scalable, adaptable, Sustainable and Foreseeable.

The distribution of key functions over a number of different entities, often known as trustees, is a typical method that is used to assure the security of voting systems [4]. Small Scale elections can be held in a centralized system, taking the role of trustees. This is in contrast to the majority of election scenarios, in which election authorities take on the role of trustees. This is the most feasible scenario for an election since Belief is evenly shared among all candidates.

[5] The usage of online voting is expanding at a rapid rate in Canada, with most of the activity centered on union elections, leadership contests for political parties, and local elections. The federal structure of the state of Canada can, at times, make the use of the internet to vote easier, while at other times it can make its usage more difficult. While taking a one-time course on blockchain technology, students at the University of Seattle created multiple voting systems.

The original (re-use) system forked Ethereum for the sake of secrecy and security. In the second method, called "re-invent," two blockchains were developed: one for validating voters, and another for protecting their ballots [6].

The election mechanism is delineated by a set of smart contracts that has to be deployed on the blockchain and managed by relevant authorities. Every voting precinct has its own blockchain-based election smart contract. E-Voting with Blockchain: A Decentralized, Private, and Secure System for Voting was written by Freya Sheer Hardwick, Apostolos Gioulis, Raja Naeem Akram, and Konstantinos Marcantonio. Commercial protocols like Bit congress, Follow My Vote, and TIVI are utilized as the basis for electronic voting in this system. The paper "Securing e-voting based on blockchain in P2P network," by Haibo Yi [7].

Academic papers in Lieu of e-voting system [8] have increased over the past two years, despite a decline in the number of nations that have implemented it. There is currently no well-developed storyline in the literature explaining the development of study into the exasperation of electronic voting. The purpose of this essay is to examine the evolution of scholarship on the topic of the widespread use of electronic voting during the past 15 years.

From 2005 to 2020, 78 studies are reviewed in this article using a semi-systematic approach. [9]Electronic voting, or "E-Voting," is a more up-to-date alternative to the use of paper ballots in elections [11,12,13,14,16].

The suggested system's architecture must enable dependability and security in order to facilitate the widespread use of electronic voting. We propose leveraging the Hyperledger Sawtooth blockchain platform to create a trustworthy, open, and distributed online voting system.

Votes are cast and recorded in an immutable blockchain state, and access to the system is restricted to election polling sites. E-Voting, often known as electronic voting, is a system that allows elections to be held without the need of paper ballots. In order for e-voting to be used on a massive scale, concerns about the safety and dependability of the system need to be addressed.

In 2008, Satoshi Nakamoto [10] announced the Blockchain technology underlying the cryptographic currency Bitcoin. This technology enables the construction and development of a safe, transparent, and decentralized system [15,17,18,19,20] in which no centralized authority has access to or influence over the voting process[21,22].

III. PROPOSED SYSTEM

Electronic voting is one area in which blockchain has a drastic impact. Due to the high level of impediments, e-voting as an individual system is not a feasible option. If an E-voting system is compromised, the repercussions will be severe. Since a blockchain network is complete, centralized its design ensures that fraud is theoretically impossible until it is adequately implemented. There is nothing coherent with blockchain technology that precludes its application to other cryptocurrencies. The concept of using blockchain technology to create a foolproof e-voting network is gaining traction. A blockchain-based voting system would be indistinguishable to end users from a conventional electronic voting system to cast votes.

Voting on the blockchain will have an encrypted piece of data that is completely public and stored on a spread-up blockchain network as opposed to a single server. If you want a vote counted quickly and accurately, use our online voting system built on the Ethereum Blockchain. Voters are able to cast their ballots from any computer with access to the internet. Data is safe from harm because of the system's secure blockchain design and several levels of protection. By leveraging blockchain technology, we can have a more trustworthy voting process that is also more secure, transparent, and unchangeable. In order to cast a ballot, voters must input their personal information. Following encryption, the data is recorded as a financial transaction. Every node in the network then receives a copy of this transaction and checks it for validity. With consensus, the transaction is recorded in a block and added to the chain.

A block that has been added to the chain cannot be removed or altered after it has been added. The outcomes are now visible, and users may go back through their previous transactions if they so want. We need to create a voting system that combines security, convenience, and confidence since existing voting methods cannot keep up with the security needs of the present age. Because of this, Blockchain technology is being incorporated into voting systems to increase security, promote voting independence from time and place, and shorten the duration and cost of the voting process. Also, the scalability of the proposed system is due in large part to the web application that is built using the React framework. Our project's loading time and overall

performance have both been improved because of the use of React JS. The system flow is shown in Fig.1.

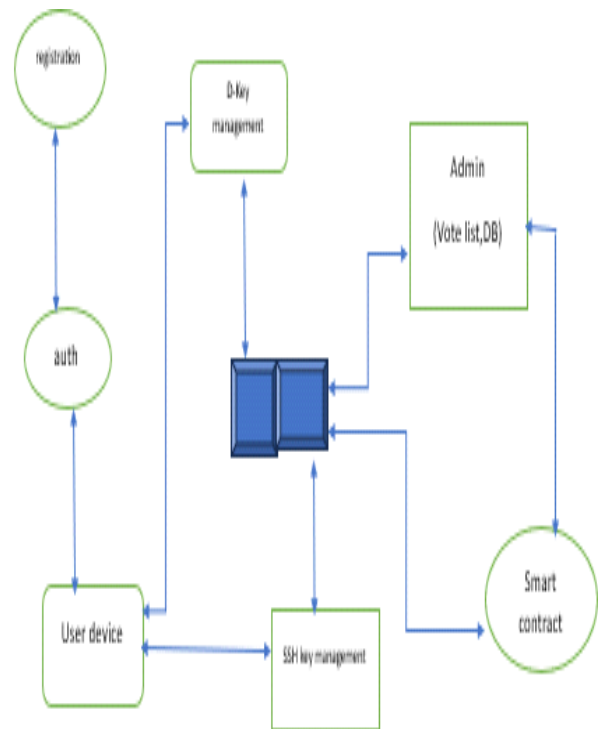


Fig.1 System flow diagram

The project's blueprints are laid out in detail. in Fig. 2. The user is required to log in using their credentials, they can register new voters after verifying their details. If the verification is unsuccessful then the application is rejected. Some of the functions are to cast votes, count the votes, add the candidates, and register and verify the voters. All these are performed by using the Ethereum server. The construction of a blockchain system, although necessary, is not enough. It has to be decentralized so that if one server goes down or a problem arises on one node, the rest of the nodes can keep running without interruption.

A. Description of the Proposed System

i. Login

Web3 apps replace the need for a username and password for authentication with a wallet-based system. The private key is used for authentication, while the public key is used to uniquely identify the account holder. Our voting web app makes use of the meta mask wallet, a plug-in for many web browsers. In order to access the app, the user needs just link it to a payment account stored in the user's digital wallet. The suggested system is designed such that each voter needs only one login. The key features of this system are its privacy protections, its one-vote-per-mobile-phone-number voting policy, and its commitment to data accuracy and openness.

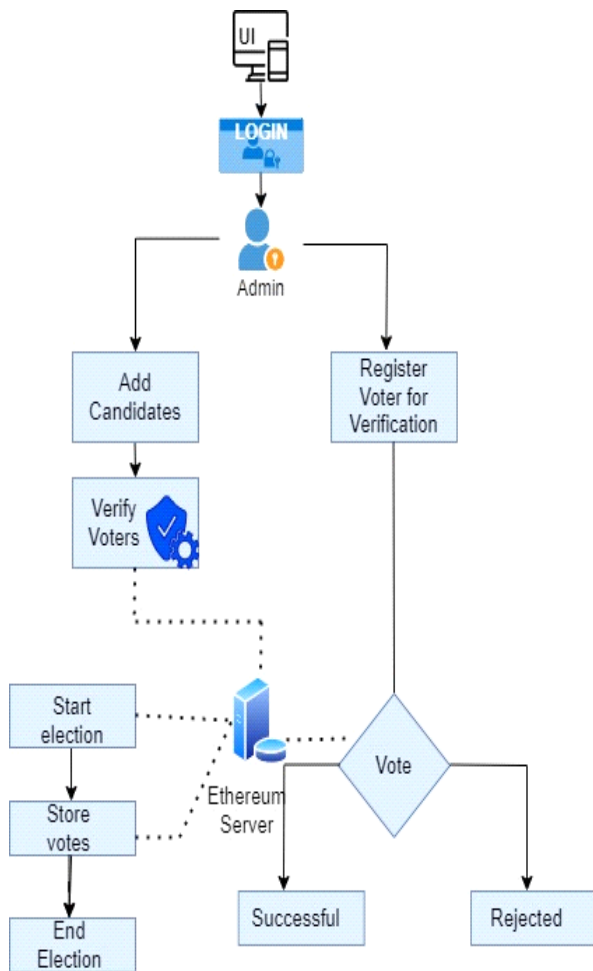


Fig.2 Architecture Diagram

ii. Initiating a new ballot

A new ballot may be initiated by any user by providing ballot details and authorizing the transaction using the wallet. With the blockchain network's approval, the transaction is added to the permanent record. The person making the ballot must include a list of valid addresses i.e., the public addresses of user accounts.

iii. Voting process

Users who are of voting age can do the same thing using a meta mask to link their accounts. After a search for a matching voter, a list of candidates that are up for election will be displayed, along with the opportunity to vote against each of them. If, however, there is no successful match, access will be terminated immediately. When a user approves a transaction in their wallet, their vote is added to the distributed ledger at the same time the transaction is broadcast to the network. To this aim, each valid vote counts as a separate transaction on the voting app's blockchain.

IV. RESULTS AND DISCUSSION

Users can see the results of the vote once the voting period has ended. As soon as a vote is cast, it is stored in the blockchain as an introductory block and in the database's

backend data tables. The system guarantees simply the "one person, one vote" democracy feature of electronic voting systems. It was determined after testing that the transaction hash representing the vote transfer was produced at the time the asset was sent to the address. Therefore, the E-voting system should be implemented initially in limited pilot areas before venturing into large-scale capacity. There are still numerous security flaws in the net and voting machines. E-voting over a secure and reliable internet will require significant security enhancements. Despite appearing to be an ideal solution, these flaws prevented the blockchain system from resolving all of the problems with the voting system. This study revealed that blockchain systems presented challenges that needed to be addressed and that there are still numerous technical obstacles. As an electronic voting option, it is important to recognize that blockchain technology is still in the budding phase. The comparison of existing and proposed technologies used for implementation are shown in Fig.3 and Fig.4.

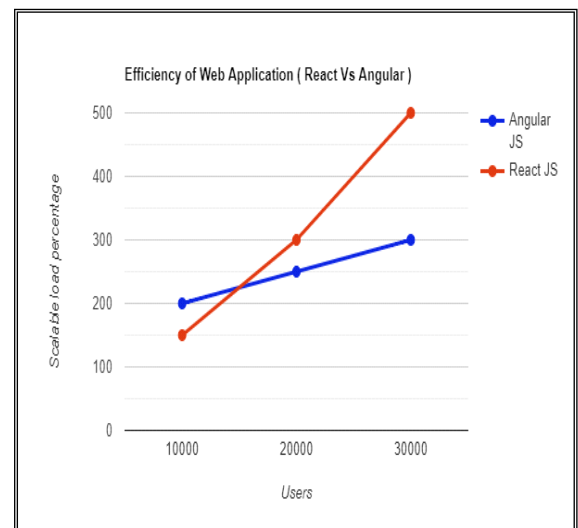


Fig.3 Comparison of React and Angular

From fig.3 it is observed that React Js is better in frontend because of its dynamic reusable components and its compatibility of creating dynamic web applications easier.

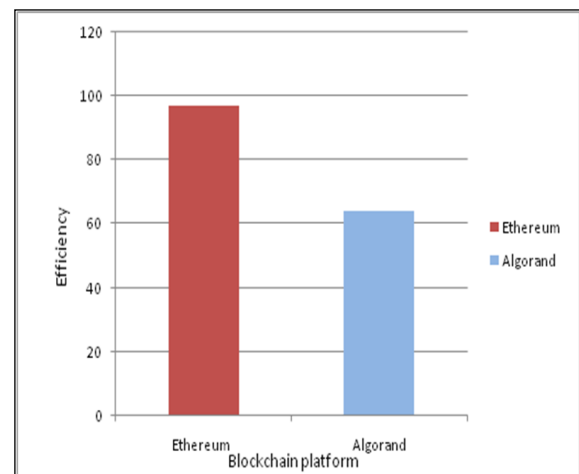


Fig.4 Comparison of Ethereum and Algorand

The above figure shows the efficiency of a blockchain application when an Ethereum algorithm is used. The main advantage of using Ethereum is it ensures that the decentralized applications and smart contracts built on its blockchain function as intended, without the possibility of fraud, interruption, or external meddling.

V. CONCLUSION

The implementation accustomed blockchain as a network-based E-voting system. This system will inhabit blockchain as an integrated system as well as a database to accumulate and hoard voters' data or accreditation which is going to harness for their authentication. The system will be using candidates' or voters' details for the voting process. Smart contracts will be accounted as responsible for managing voting procedures and results. Our system boosts the efficacy of the authentication and of the conferring of a candidate's vote. Blockchain technology encrypts the vote, preventing every vote from being tampered with. It ensures that a voter can only vote for a candidate once. The system quickly retrieves election results, thereby reducing labor costs and counting errors.

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